

Source data and preservation

Foundational documentation for all twelve analysis maps

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Source lineage

Publisher context: <https://human-settlement.emergency.copernicus.eu/ucdb2024Overview.php> and https://human-settlement.emergency.copernicus.eu/ghs_ucdb_2024.php. The JRC overview describes 11,422 urban centres and integrated thematic indicators. Local exports are labelled R2024A; their precise minor version is not proven by the current website release. Source reference years can include modelled estimates and are not direct contemporary surveys.

The original ArcGIS projects and geodatabases were inspected, along with the CorrectGeo project and recovered final notebook analyses from Global Human Settlements Setup. The source manifest records exact resolved dataset paths; similarly named layers are not sufficient proof of identity. Full foundations were selected before analysis so ecclesiastical fields remained available.

Local foundation	Features	Original attributes
Dioceses	2874	43
Conferences	122	186
Members	2739	92
Members20km	2739	13
Members100km	2739	13
Earlier GHS UCDB	13135	53
Climate R2024A	11422	125
Water R2024A	11422	44
Infrastructure R2024A	11422	150
Socioeconomic R2024A	11422	183
Greenness R2024A	11422	129
Hazards R2024A	11422	225
Health R2024A	11422	84

The 13,135-feature earlier GHS_UCDB reference and the 11,422-feature R2024A thematic city universe are kept separate. There is no cross-release city-ID join. The seven thematic R2024A tables share identical unique ID_UC_G0 sets.

Preserving ecclesiastical and member data

All 43 diocesan and 186 conference original attributes were preserved and compared value-for-value. Names, rites, traditions, hierarchy, source dates, population, Catholic statistics, clergy, parishes, leadership, enrolment, education and duplicate source series remain available. Member and buffer source attributes and geometry also remain. GS_Ecclesial provides readable context in longer summaries. Leadership and historic statistics retain source dates; they were not revalidated as current officeholders. Fields with suffix _26 and other original series were not merged or silently overwritten.

The manifest identifies the resolved diocesan source

DiocesesoftheCatholicChurchv2CatholicStats_SummarizeWithin and conference foundation copy. It is the selected full source, rather than an assumed field-stripped layer merely sharing the requested display name. Consult Source_Dataset_Inventory.csv and Evidence/foundation_manifest.json for exact paths.

Spatial preparation

ExportFeatures copied original attributes and geometry; original object identifiers were retained as GS_SourceID. Slim temporary joining copies were projected to WGS 84 EPSG 4326. JOIN_ONE_TO_MANY, KEEP_COMMON, INTERSECT relationships retained explicit GS_Left and GS_Right keys to avoid field-name collisions. Unique pairs were checked. Analysis display uses Equal Earth EPSG 8857.

Relationship	Unique pairs
Member point to city	1,929
City to diocese	12,395
City to conference	11,529
20 km buffer to city	4,499
100 km buffer to city	26,929
Member point to diocese	2,670
Member point to conference	2,670

Cities use whole mapped urban centres. Members inherit values only after one unique city match: 1,929 of 2,739 point records match 638 cities; 810 unmatched records keep null indicators and zero city-linked counts. Dioceses, conferences and member buffers use an equal-city mean of distinct intersecting cities, separately for each metric's valid observations. These are neither population-weighted nor area-weighted territorial estimates. A cross-boundary city can occur in multiple summaries, so totals across overlapping territories are not additive.

622 dioceses and two conferences have no intersecting mapped city; 367 twenty-kilometre buffers and 73 hundred-kilometre buffers have none. Empty city context stays unavailable rather than inventing a zero environmental value.

Metric source field crosswalk

Derived metric	Source or calculation
GS_Pop15	Infrastructure / GH_POP_TOT_2015
GS_Pop25	Infrastructure / GH_POP_TOT_2025
GS_PopGrowth	$100 \times (\text{Pop25} - \text{Pop15}) / \text{Pop15}$
GS_Built15	Infrastructure / GH_BUS_TOT_2015 ÷ 10,000
GS_Built25	Infrastructure / GH_BUS_TOT_2025 ÷ 10,000
GS_BuiltGrowth	$100 \times (\text{Built25} - \text{Built15}) / \text{Built15}$
GS_GrowthGap	BuiltGrowth – PopGrowth
GS_BuiltChange	Built25 – Built15
GS_GreenPop	Greenness / SD_POP_HGR_2025
GS_HeatDays	Climate / CL.UTC_T32_2020
GS_FloodPop	Hazards / EX_100_SHP_2025
GS_Groundwater	Water / NS_ESB_GWA_2025
GS_Hospital	Health / HL_SHP_HOS_2025
GS_Pharmacy	Health / HL_SHP_PHA_2025
GS_PM25	Health / EM_PM2_CON_2020
GS_Education	Infrastructure / IN_CIS_EDU_2020
GS_Canopy	Greenness / GR_CTH_AVG_2020
GS_CanopySD	Greenness / GR_CTH_STD_2020
GS_GreenIndex20	Greenness / GR_AVG_GRN_2020
GS_GreenCover20	Greenness / GR_SHB_GRN_2020
GS_GreenIndex25	Greenness / GR_AVG_GRN_2025
GS_GreenCover25	Greenness / GR_SHB_GRN_2025
GS_GreenChange	GreenCover25 – GreenCover20
GS_HazardTypes	Hazards / count of positive HZ_CEV_*_2015 fields
GS_Crop15	Greenness / LU_HEC_040_2015
GS_Crop19	Greenness / LU_HEC_040_2019
GS_CropChange	Crop19 – Crop15
GS_CropGrowth	$100 \times (\text{Crop19} - \text{Crop15}) / \text{Crop15}$

The seven thematic export names, original aliases, spatial references and field types are in `Original_Source_Inventory.json`. Full final aliases, types, text capacities and original-field flags are in `Field_Dictionary.csv`. `Source_Dataset_Inventory.csv` is a compact path and count index; `Preserved_Source_Fields.csv` enumerates unchanged source attributes.

Quality control register

Evidence file	What it establishes
foundation_manifest.json and analysis_qc.json	Copies, counts, preserved fields, read-back calculations and foundation checks
relationship_qc.json	Explicit pair counts and matching scope
independent_validation.json	Specified prior-final comparisons and independent mean/rank checks
reach_qc.json	100 km reach totals and boundary exclusions
reach_documentation_qc.json	New nine-field reach check against saved relationships
member_ecclesiastical_qc.json	Readable member-location ecclesiastical enrichment
unmatched_member_context_qc.json	810 unmatched records retain null city indicators and zero city-linked counts
final_project_validation.json	15 maps, 12 analyses, no broken layers and 88 saved popup checks

Saved calculations and valid city counts were checked after writing. All 28 metric rankings were independently verified for cities, dioceses, conferences, 20 km buffers and 100 km buffers. Aggregate means were reproduced from saved relationship keys. Original geometry and attributes were compared for the five member and ecclesiastical foundations. This validates the implemented calculations, not the accuracy of every publisher indicator or a new independent spatial-overlay algorithm.

QC is computational and structural. It does not establish current leadership, perfect city delineations, reporting completeness, causality, intervention effectiveness or address-level environmental exposure. Source field availability is distinct from scientific data coverage.

Cleaning and numeric interpretation

Null, nonnumeric, nonfinite and sentinel values ≤ -999 are excluded. Negative values are excluded for nonnegative measures, but signed indices and changes can remain negative. Percentages must be within 0–100; heat days within 0–366. Percentage changes require a positive baseline. Compound screening uses three-valued logic: AND is false if any component is false, true if all are true, otherwise unknown; OR is true if any is true, false if all are false, otherwise unknown.

Rank 1 is the highest numeric value, with competition ranks for ties. Percentile is $100 \times (\text{number below} + \text{half the number equal}) / \text{valid N}$. Higher percentile means a higher numeric value, not better or worse conditions. Missing values are excluded. All-equal values have rank 1 and percentile 50.

Workflow record

Work completed: inspect source projects and notebooks; select and copy complete foundations; preserve source identifiers, attributes and geometry; build explicit spatial relationships; calculate 28 metrics and thematic screens; summarize six spatial contexts; calculate separate ranks and percentiles; reconstruct 100 km reach; build twelve themed maps plus source, foundation and CorrectGeo reference maps; add aliases, short popups and longer sidebar text; enrich ecclesiastical identity; repair unmatched-member city counts; validate calculations, prior comparisons and saved project connections; export the field dictionary, CSVs, guide and overview graphic; add this fourteen-document package and a new read-only reach audit.

Workflow/ contains the actual analysis and documentation-check code archived for review.

Earlier_Setup_Notebooks/ archives the recovered prior setup cells for historical comparison; the current Workflow is the implementation used for this project. The original current scripts remain in GHSL. Execution needs ArcGIS Pro and arcpy, the original datasets and current paths. The final CSV exports and Global_Human_Settlements.gdb remain in GHSL rather than being duplicated in every documentation folder. The existing guide and overview graphic are also copied here for reference.

Source and delivery dependencies

Main project: ../.. / LSAP_Global_Human_Settlements.aprx. Main analysis database: ../.. / Global_Human_Settlements.gdb. Original CorrectGeo reference layers still link to local CorrectGeo sources. Country context uses the local BNDA_simplified shapefile. Moving the documentation alone does not package those GIS dependencies.

Scientific use

Use these analyses as descriptive city-context screens for Laudato Si dialogue, local verification and partnerships. Population and built-up change are not project-level causes; hospital proximity is not access or quality; environmental concentrations are not individual health diagnoses; a 100-year flood is a return-period indicator; canopy height is not coverage or biodiversity; positive groundwater transgression is not proof of a particular mechanism; hazard zeroes are not evidence of safety; cropland loss alongside later growth is not proof of conversion or food insecurity.